WORKED SOLUTION

STEP I 2007, Question 8

Pure • Curve sketching

STEP I 2007, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s1-q8

The key idea

Two curves touch at a point exactly when the equation for their intersections has a repeated root there. So look at the difference between the two cubics and expect a factor $(x-2)^2$. The coefficients $1, -6, 12, -8$ in the a-terms are the binomial coefficients of $(x-2)^3$, and spotting this makes both the factorisation and the sketches much easier.

Solution

Touching at $(2,8)$ and the other intersection

Grouping the terms that contain a,

$$ax^3 - 6ax^2 + 12ax - 8a = a(x-2)^3, \qquad \text{so} \qquad y = a(x-2)^3 + 12x - 16. \qquad (*)$$

The intersections with $y = x^3$ are the roots of

$$a(x-2)^3 + 12x - 16 - x^3 = 0.$$

Now $x^3 - 12x + 16$ vanishes at $x = 2$, and dividing out gives $x^3 - 12x + 16 = (x-2)^2(x+4)$. Hence

$$\begin{aligned} a(x-2)^3 + 12x - 16 - x^3 &= a(x-2)^3 - (x-2)^2(x+4) \\ &= (x-2)^2[(a-1)x - 2a - 4]. \end{aligned}$$

At $x = 2$ both curves pass through $(2,8)$, and the factor $(x-2)^2$ shows that $x = 2$ is a repeated root of the intersection equation. Equivalently, the gradients agree: from $(*)$, $y' = 3a(x-2)^2 + 12$, which is 12 at $x = 2$, and the gradient of $y = x^3$ there is $3 \cdot 2^2 = 12$. So the curves touch at $(2,8)$, for every value of a.

Any other intersection comes from $(a-1)x = 2a + 4$.

- If $a \neq 1$, then $x = \dfrac{2a+4}{a-1}$, which is never 2 (that would need $2a + 4 = 2a - 2$). So

there is exactly one other point,

$$\left(\frac{2a+4}{a-1}, \left(\frac{2a+4}{a-1}\right)^3\right).$$

- If $a = 1$, the bracket is the constant -6, so there is no other point of intersection.

Ask yourself Which curve is on top, and where?

The factorisation answers this for free. Since $(x-2)^2 \geq 0$, the sign of $(*)$ minus x^3 is the sign of the linear factor $(a-1)x - 2a - 4$. That tells us on which side of the other intersection the curve $(*)$ lies above $y = x^3$, which is most of what a sketch needs.

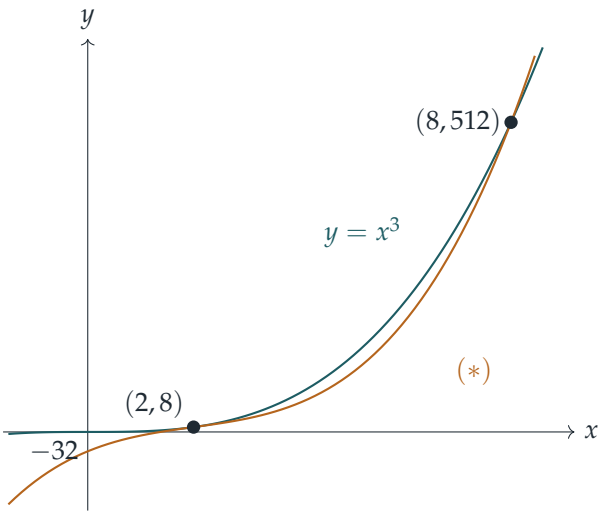
In all three sketches, $(*)$ has a point of inflection at $(2, 8)$ with gradient 12, because $y'' = 6a(x-2)$ changes sign there when $a \neq 0$. So $(*)$ and $y = x^3$ share the tangent $y = 12x - 16$ at $(2, 8)$; $y = x^3$ is concave up there while $(*)$ is not.

Part (i): $a = 2$

Here $y = 2(x-2)^3 + 12x - 16$ and

$$y - x^3 = (x-2)^2(x-8).$$

The other intersection is $(8, 512)$. The curve $(*)$ lies below $y = x^3$ for $x < 8$ (touching at $x = 2$) and above it for $x > 8$. Its gradient $6(x-2)^2 + 12$ is always positive, so it is increasing with no turning points. It meets the y-axis at $(0, -32)$ and crosses the x-axis once, near $x \approx 1.37$.



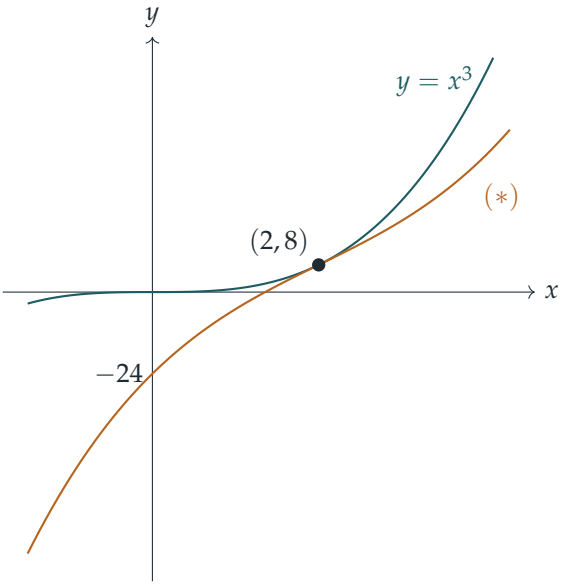
(The vertical scale is heavily compressed, so the touching at $(2, 8)$ looks flat; the two curves share the tangent there.)

Part (ii): $a = 1$

Here $y = (x-2)^3 + 12x - 16 = x^3 - 6x^2 + 24x - 24$, and

$$y - x^3 = -6(x-2)^2 \leq 0.$$

So $(*)$ lies below $y = x^3$ everywhere except at $(2, 8)$, where it touches; there is no other intersection. The gap $6(x-2)^2$ grows as we move away from $x = 2$ in either direction. The gradient $3(x-2)^2 + 12$ is always positive, so $(*)$ is increasing. It meets the y-axis at $(0, -24)$ and crosses the x-axis once, near $x \approx 1.36$.



Part (iii): $a = -2$

Now the other intersection has $x = \dfrac{-4+4}{-3} = 0$, so it is the origin $(0, 0)$, and

$$y - x^3 = (x-2)^2(-3x) = -3x(x-2)^2.$$

So $(*)$ is above $y = x^3$ for $x < 0$ and below it for $x > 0$, touching at $(2, 8)$.

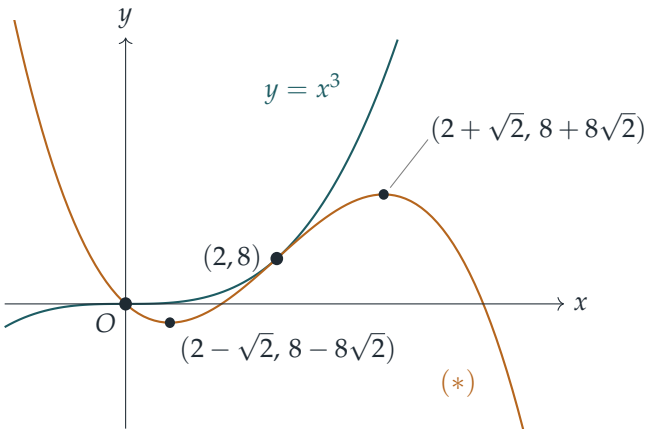
Expanding, $y = -2(x-2)^3 + 12x - 16 = -2x^3 + 12x^2 - 12x = -2x(x^2 - 6x + 6)$, which crosses the x-axis at $x = 0$ and $x = 3 \pm \sqrt{3}$ (about 1.27 and 4.73). The gradient is

$$y' = -6(x-2)^2 + 12,$$

which is zero when $(x-2)^2 = 2$, that is, $x = 2 \pm \sqrt{2}$. Substituting,

$$x = 2 - \sqrt{2}: \quad y = 8 - 8\sqrt{2} \approx -3.31 \quad \text{(minimum)},$$
$$x = 2 + \sqrt{2}: \quad y = 8 + 8\sqrt{2} \approx 19.3 \quad \text{(maximum)}.$$

Since the leading coefficient is negative, $y \to +\infty$ as $x \to -\infty$ and $y \to -\infty$ as $x \to +\infty$. At the origin $(*)$ has gradient -12, while $y = x^3$ has gradient 0, so the curves genuinely cross there.



Checks

- At $x = 2$, $(*)$ gives $a \cdot 0 + 24 - 16 = 8$ for every a, and $y' = 12$, matching $y = x^3$.
- With $a = 2$ and $x = 8$, the neat form gives $2 \cdot 6^3 + 96 - 16 = 512 = 8^3$, confirming the point $(8, 512)$.
- With $a = 0$, $(*)$ is the tangent line $y = 12x - 16$, and the formula gives the other intersection at $x = -4$; indeed $x^3 - 12x + 16 = (x-2)^2(x+4)$.
- We checked the factorisation exactly for thousands of random rational a and x, and the turning values $8 \pm 8\sqrt{2} \approx 19.31,\ -3.31$ in part (iii) numerically.

Common slips

- Forgetting the case $a = 1$, where the formula for the other intersection breaks down because there isn't one.
- Showing only that both curves pass through $(2, 8)$. Touching also needs equal gradients, or equivalently a repeated root.
- In part (iii), drawing the curves as touching at the origin. They cross there, with different gradients.

Beyond STEP

A repeated root of $f(x) - g(x)$ is the algebraic shadow of tangency: the curves agree to first order at that point. Pushing this further, the order of contact counts how many derivatives agree, and it is the idea behind Taylor polynomials, which are the

polynomials with the highest possible order of contact with a function at a point. Bartle and Sherbert, *Introduction to Real Analysis*, develops Taylor's theorem carefully. Siklos, *Advanced Problems in Mathematics*, has more curve sketching in the STEP style.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.